Problem Set 4

QBCS 5002: Introduction to Quantitative Biology

Due: November 17, 2026, at 12:00 p.m.

The problem set is to be completed by teams of two. We strongly encourage teams pairing a student from a biology, medicine or chemistry background with a student from a mathematics, physics or engineering background. Each team will submit one joint solution, and both partners will receive the same grade.

Your solutions should show the main algebraic steps. For the coding problems, you may use Python, Mathematica, MATLAB, C/C++, or another language approved by the instructor. Submit readable source code with comments identifying the key mathematical and numerical steps. The complete submission should include a PDF report with commented source code in a separate file or as appendix. Cite the paper and external libraries or code resources used.

Problem 1. From Bursty Protein Production to Stationary Distributions

Reference: L. Cai, N. Friedman, and X. S. Xie, "Stochastic protein expression in individual cells at the single molecule level," *Nature* **440**, 358–362 (2006).

Cai *et al.* observed that protein production at low expression levels occurs in stochastic bursts, with the number of molecules per burst following an exponential distribution. The expression of proteins from a given gene can be characterized by two parameters: the average number of bursts per cell cycle, a, and the average number of proteins per burst, b. At steady-state, protein production is balanced by protein dilution due to cell growth and division.

1. Prove that the steady-state probability distribution of protein abundance x follows a Gamma distribution
$$p(x) = \frac{x^{a-1}e^{-x/b}}{b^a\Gamma(a)}$$

2. Calculate the mean and variance of the stationary distribution and show how they are related to the parameters a and b. Interpret the biological meaning of a and b.

Problem 2. From Chemical Master Equation to Gillespie Simulation

Reference: N. D. Lord *et al.*, "Stochastic antagonism between two proteins governs a bacterial cell fate switch," *Science* **366**, 116–120 (2019). doi:10.1126/science.aaw4506.

Cell fate decisions reflect a balance between stochasticity and determinism: genetically identical cells can adopt a multitude of fates, while these states remain distinct, stably maintained, and coordinated with neighboring cells. Such switching is often associated with regulatory feedback and deterministic bistability. However, in the 2019 *Science* paper, Lord *et al.* showed

that the switching of *Bacillus subtilis* between a motile, unicellular state and a chained, multicellular state can instead arise from stochastic competition between two constitutively produced proteins: One is SinR (R), a transcriptional repressor of genes associated with the chained state, and the other is its antagonist SinI (I), which binds SinR to form an inactive complex.

In the simple stochastic competition model, I and R are produced with rates λ_I and λ_R, respectively, form an inactive complex with rate cIR, and are removed by dilution or degradation with a characteristic lifetime τ. A SinR-repressed downstream target serves as a reporter of the regulatory state. I and R are assumed to be produced one at a time (i.e., $b = 1$). A simple model of the SinI-SinR interaction can be represented by

$$\begin{aligned} I &\xrightarrow{\lambda_I} I+1, \qquad & I &\xrightarrow{I/\tau} I-1 \\ R &\xrightarrow{\lambda_R} R+1, \qquad & R &\xrightarrow{R/\tau} R-1 \\ (R,I) &\xrightarrow{cRI} (R-1, I-1) \end{aligned} \tag{1}$$

Because the abundances of I and R fluctuate stochastically, rapid sequestration can convert small relative differences between them into alternating SinR-dominant and SinI-dominant states.

1. Using the transition rates above, derive the corresponding Chemical Master Equation for the joint probability
$$P_{i,r}(t) = P\{I(t) = i,\ R(t) = r\}.$$

2. Implement Gillespie's stochastic simulation algorithm and generate representative trajectories of $I(t)$ and $R(t)$. Use $\tau = 1, \lambda_R = 10, c = 5$, and compare
$$\frac{\lambda_I}{\lambda_R} = 0.6,\ 0.7,\ 0.8$$
starting from $I(0) = R(0) = 0$. Explore how the stochastic dynamics and the occurrence of SinR-dominant and SinI-dominant episodes change as λ_I/λ_R increases.

3. Derive the corresponding deterministic mass-action ODEs and simulate them using the same kinetic parameters. Determine the steady state and examine its stability.

4. Compare the stochastic and deterministic dynamics. Explain how stochastic fluctuations and rapid SinI–SinR sequestration can generate transitions between SinR-dominant and SinI-dominant states even though the deterministic system has only one stable steady state. Why is this behavior different from classical deterministic bistability?

* For tasks 2 and 3, submit your commented code, figures with clearly labeled axes and units, and a brief interpretation of the results.

Problem 3. Noise and Excitability in the MeKS Module

Reference: G. M. Süel et al., "An excitable gene regulatory circuit induces transient cellular differentiation," *Nature* **440**, 545–550 (2006), and its Supplementary Information.

The MeKS module regulates transient competence in *Bacillus subtilis*. ComK activates its own expression and indirectly represses ComS expression, while ComS competitively inhibits MecA-mediated degradation of ComK. The reduced deterministic model is

$$\frac{dK}{dt} = a_k + \frac{b_k K^n}{k_0^n + K^n} - \frac{K}{1 + K + S} \tag{1}$$

$$\frac{dS}{dt} = \frac{b_s}{1 + (K/k_1)^p} - \frac{S}{1 + K + S} \tag{2}$$

In the equations above, K and S are dimensionless protein concentrations rescaled with Γ_k and Γ_s, and t is dimensionless time rescaled with δ: $K = [\text{ComK}]/\Gamma_k, S = [\text{ComS}]/\Gamma_s, t = \delta t_{\text{phys}}$.

Assume a well-mixed cell of fixed volume $V = 4\,\mu\text{m}^3$, approximating a cell of dimensions $4 \times 1 \times 1\,\mu\text{m}$, with $\delta_k = \delta_s = \delta = 0.0014\,\text{s}^{-1}, \Gamma_k = 500\,\text{nM}, \Gamma_s = 50\,\text{nM}$.

Tasks

1. **Deterministic dynamics and excitability.** Using a representative phase portrait in the excitable regime, such as Fig. S3D in the Supplementary Information, sketch or plot the vector field and identify the vegetative state, the saddle point, and the unstable spiral. Illustrate how a sufficiently large perturbation from the vegetative state can trigger a large excursion through the competence region before the system returns to the vegetative state. A clear hand-drawn phase portrait or a computational plot is acceptable.

   For the following tasks, use the parameters corresponding to Fig. S3A in the Supplementary Information: $a_k = 0.004, b_k = 0.08, b_s = 0.68, k_0 = 0.2, k_1 = 0.222, n = 2, p = 5$.

2. **Chemical Master Equation.** Construct an effective stochastic model with four single-molecule reaction channels: production and removal of ComK, and production and removal of ComS. Let n_K and n_S denote the corresponding molecule numbers. Write the four elementary reactions and their propensity functions $R_K^{\pm}(n_K, n_S)$ and $R_S^{\pm}(n_K, n_S)$. Derive the Chemical Master Equation for the joint probability distribution of n_K and n_S.

3. **Langevin equation and Linear Noise Approximation.** Derive the Langevin equation corresponding to the four reaction channels. Linearize the dynamics around the vegetative fixed point and obtain the Linear Noise Approximation for the molecule-number fluctuations.

4. **Stationary covariance and noise**. Calculate the stationary covariance matrix σ. Write the corresponding Gaussian distribution, and calculate the relative variances of n_K, n_S, together with the correlation coefficient between ComK and ComS fluctuations.

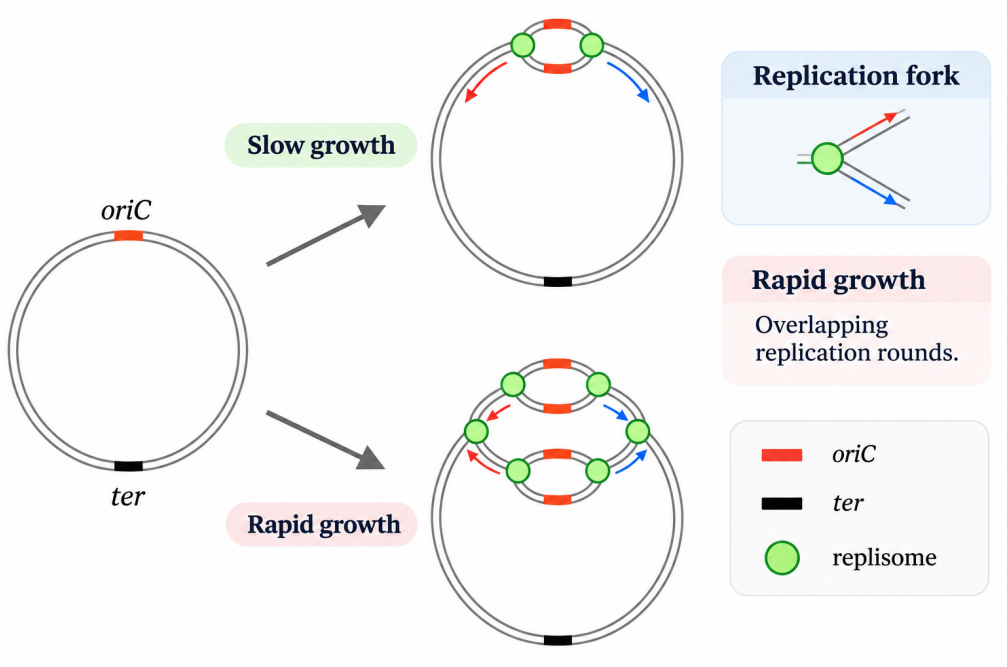
Problem 4. Intrinsic and Extrinsic Noise in Gene Expression

Reference: M. B. Elowitz, A. J. Levine, E. D. Siggia, and P. S. Swain, "Stochastic Gene Expression in a Single Cell," *Science* **297**, 1183–1186 (2002).

Gene expression varies substantially among genetically identical cells. In bacteria such as *E. coli*, chromosome replication initiates at the origin of replication (*oriC*) and proceeds

bidirectionally along two replication forks. Because different genomic loci are copied at different times, gene copy number can depend on chromosomal position. During rapid growth, new rounds of replication may begin before earlier rounds are completed, leading to a higher average gene dosage near the replication origin than near the terminus.

Elowitz *et al.* used two fluorescent reporters, CFP and YFP, controlled by identical regulatory sequences in the same *E. coli* cell to distinguish intrinsic and extrinsic noise. The two reporter genes were integrated at chromosomal loci approximately equidistant from and on opposite sides of the origin of replication, minimizing systematic differences in gene copy number while retaining sensitivity to stochastic differences.



Task: Deletion of *recA* was found to substantially increase intrinsic noise. Since RecA helps rescue stalled replication forks, explain how loss of RecA may increase intrinsic noise through transient copy-number differences between the two reporter genes.